

transform the way medical records are maintained. The open and secure nature of the blockchain ensures the safety of an individual's medical records. It does away with fragmentation and redundancy, and makes the record-keeping transparent and efficient. It also prevents lock-in. When you want to consult a specialist from across the world, you do not have to wait for your current physician to give you a copy of your records. You can retain control over your records and give access to a trusted individual at the click of a button! Chroni-cled, Ever, and Patientory are some of the blockchain based solutions for medical data management. EMR-Chain and Plenum Data are some of the homegrown solutions.

Product design: Defy the laws of the physical world

The field of product design is in for a mega upgrade in the metaverse era. There are three aspects to this: improvement in the design process, creation of new products for the metaverse, and metaverse integration in traditional physical products.

When you design a product for the metaverse, you can let your imagination run wild, and can create products that can otherwise exist only in dreams! The metaverse does not follow the laws of physics, so you can create drastically different products and experiences. For example, you can create a house on Mars, shoes that make you fly, or a car that hops!

You can also integrate the metaverse into a real, physical object. For example, an immersive lounge chair, which creates a real-time view of a beach or valley of your choice. In CES 2022, LG showcased a concept car or mobile cabin called the Omnipod, a fully autonomous vehicle with an onboard living space that adapts to the passenger's unique needs. It has an expansive tunnel screen called the meta-environment screen, which stretches from the



Fig. 1: The living space of LG's Omnipod concept car adapts to the user's needs using a meta-environment screen (Courtesy: LG)

floor to the ceiling, helping to recreate a chosen environment, be it your lounge, workspace, bedroom, or garden! The concept cabin is replete with a virtual assistant called Reah, who also adapts to the chosen environment. For a spot of entertainment, you can connect the Omnipod's tunnel and multimedia screens to get a feeling of virtually entering the movie you are watching! There are exterior cameras, which let you also keep an eye on the road.

The metaverse can also drastically upgrade the very process of designing something. You can collaborate with people across the world to create cross-functional objects. You can first create 3D prototypes and test them in a virtual environment to make sure they work as intended before going to the next stage of development. Even for something as simple as a new series of totes or sneakers, you can first launch the designs as NFTs and gauge the customer response, before physically manufacturing the products. This would reduce the cost of development and risk of failure.

Learning: A long way to go

The pandemic made every student realise that learning through online platforms was not the same as being

in school with friends, and they are delighted to be back on campus. However, educationalists still worry about what would happen if another such situation arises where kids are unable to go to school. As a work-around, experts suggest that schools attempt a multi-modal education even on a normal day, so that they can quickly transition to a digital mode when required.

While schools struggled for continuity using e-learning platforms, videos, and chatrooms, in the future the metaverse could be a more effective option. An immersive learning experience with graphics-intensive virtual platforms will empower students to understand concepts better, and to experience how certain things work in the real world. It might be difficult for children to understand how a bank works. What if they could enter a simulated environment and experience it for themselves? The metaverse makes that possible—and it might be a great learning aid even on a normal day.

However, the fact remains that the metaverse is still in a very nascent stage and its constituent technologies are too expensive to have a wide reach. Where it could work wonders is in higher education, upskilling, and corporate training.